Jacob Baker-Spitzer

Durham, NC 917-940-5546 jacob.baker-spitzer@duke.edu LinkedIn GitHub Portfolio

Education

Duke University, Pratt School of Engineering **Expected May 2030**
B.S.E. Electrical & Computer Engineering; Intended Second Major: Mathematics *Durham, NC*

- **Coursework:** Linear Algebra, Computing for Engineers (Python), Engineering Design & Communication, Economics

NEST+m (New Explorations into Science, Technology + Math) **Jun 2026**
High School Diploma *New York, NY*

- Selective NYC public school for math and science; National Honor Society, Advanced Jazz Band, Baseball Club co-founder

Experience

Research Assistant **Jun 2026 – Jul 2026**
Prof. Baker, Department of Economics, Hunter College (CUNY) *New York, NY*

- Built a 1,000+ building dataset covering ∼100 years of NYC construction, later used in a published economics paper
- Extracted floors, elevators, and fireproofing for each building from historical fire records in Google Sheets over 4 weeks
- Verified lot labels against a single source of truth to keep the full database accurate and consistent for downstream analysis

New Member Education Cohort **Sep 2026 – Present**
Catalyst, Duke Pre-Professional Technology Organization *Durham, NC*

- Selected as 1 of 20 from ∼200 applicants through a multi-round recruitment process that included a technical interview
- Training across software engineering, AI/ML, data science, and quant tracks, and completing hands-on projects in each

Incoming Engineering Trainer, Light/Energy Team **Jan 2027 – May 2027**
Ignite, Duke Engineering Outreach for Durham Middle Schoolers *Durham, NC*

- Will lead weekly hands-on light and energy engineering projects for Durham middle school students in a paid spring role

Projects

Adaptive Fishing Game | *Arduino, Stepper Motors* **Aug 2026 – Present**

- Built a fully automatic, switch-controlled fishing game for a child with limited mobility as part of a 5-person team
- Took the design through 3 prototypes (low-fidelity, high-fidelity, final), with each round paced to last about 1 minute
- Drew the circuit diagram (Arduino, 2 stepper motors and drivers, fuses, limit switches) and designed the drop mechanism
- Soldered and integrated electronics as a team to turn our code into a working product, and prepared team presentations

Bronx Housing Analysis | *Python, pandas, GeoPandas* **Sep 2026 – Oct 2026**

- Mapped housing conditions in my home Bronx neighborhood (25 census tracts) using 45,736 city housing-violation records
- Found ∼30% fewer violations per renter household than the Bronx overall, then presented my findings to the Catalyst club

Bank Customer Prediction Model | *Python, scikit-learn* **Oct 2026**

- Built a Python model predicting which of ∼30,900 bank customers will sign up for a savings account after a bank sales call
- Scored 0.79 ROC-AUC, beating a guess-"no"-every-time baseline, and held up on new data the model had never seen

Leadership & Activities

Black Belt (First Degree) & Assistant Instructor **Aug 2013 – Present**
Riverdale Kenshikai Karate *Bronx, NY*

- Earned a first-degree black belt through 13 years and 600+ classes of steady training at the same dojo, starting at age 4
- Co-taught 50+ classes of 10–20 students, giving one-on-one help to ∼50 students struggling with form and technique
- Helped organize dojo tournaments of 50–60 competitors, tallying the judges' scores live in a spreadsheet during each round

Certified Youth Soccer Referee **Sep 2023 – Jun 2026**
Westchester Youth Soccer League *Westchester, NY*

- Officiated 180+ youth matches (U9–U12, 7v7 and 11v11) over 3 years, including several WYSL tournament games
- Advanced from assistant to center referee, making real-time calls and calmly managing players, coaches, and parents
- Mentored a newly certified referee through his first match, coaching him in real time on positioning, signals, and calls

Varsity Soccer & Tennis Captain **Sep 2022 – Present**
NEST+m; Duke Club Soccer *New York, NY / Durham, NC*

- Soccer: 4-year varsity, 2-year captain, NYC champion. Tennis: 2-year captain and two-time NYC championship finalist
- Ran team practices as captain and led a team fundraiser to buy new matching jerseys and other team gear for the season

Technical Skills

Languages: Python (pandas, GeoPandas, scikit-learn, matplotlib), Arduino (C/C++)
Tools: VS Code, Git/GitHub, Arduino IDE, Onshape (CAD), Google Sheets
Hardware: Stepper Motors & Drivers, Soldering, Circuit Prototyping, 3D Printing